Mobiclip FAQ

Ver 1.2.0

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Revision History

Ver.	Rev. Date	Content of Revision
1.2.0	2010-9-03	Removed chapter describing logo usage. Removed "About transitioning from VX Middleware" chapter. Added chapter on changes done in VFW codec with Mobiclip license.
1.1.1	2008-9-11	Fixed Q ‘5.2 Is it possible to do the decode processing within a separate thread’.
1.1.0	2008-5-7	Added ‘3.2 How to insert Keyframe at any position(point)’.
1.0.0	2008-4-8	First Edition

1 Introduction

1.1 About this document

This document compiles some frequently asked questions relevant to the Mobiclip Middleware. The questions have been chosen among the most frequent inquiries for the VX Middleware, the previous movie playback library. For details about the “Mobiclip Middleware”, please refer to both the ‘QuickStart.pdf’ and the Mobiclip function reference manual.

2 Video Editing and Codec

2.1 What is the method to avoid misprocessing of the movie?

Q. At a certain particular scene of the movie, misprocessing occurs. Is there any way to avoid it?

A. Fundamentally, please work around this by lowering the encoding quality.

If it is limited to a certain particular scene, there is a way to cut the movie into separate segments then splice them together. By encoding the particular scene segment at a lower quality, you can reduce the required CPU resource and avoid the misprocessing.

As a reminder, the difference of picture quality or color at the splice point sometimes becomes visible with this method. So please pay special attention if you intend to use this method.

2.2 How to insert a Keyframe on specific positions.

Q. Although I tried to force Keyframes at certain positions in my movie, setting up the codec adequately, no keyframes were output where I desired.

A. Adjusting the value of Keyframe Boost, Detection threshold and Maximum Interval during the encoding will affect the settings of the keyframe insertion. But the actual location of key frames varies on the movie content itself. For more detailed explanation of these parameters, please refer to the “QuickStart.pdf”.

If you really want to force a key frame at a specific point, we recommend you to split your movies and save them so that each chunk start is a frame you want to be a key frame. Then encode each chunk separately and splice them back together. Using this method, you can insert a key frame exactly where you want.

2.3 What is the return value of AviToMobiclipDs.exe?

Q. We are planning to use a batch file to execute AviToMobiclipDS.exe, and if any error occurs we can change the process. For this purpose, we want to use a Return Value. What are the return values of AviToMobiclipDs.exe?

A. The return value is as follows:

Normal end：0

Abnormal end：1

2.4 Noise in sound data occurs when the movie has been converted.

Q. When we checked the data converted with AviToMobiclipDs.exe, there was some noise in the sound data. Even though we set the –quality option to the maximum level(1000) at the time of conversion, it did not improve. What else can we do?

A. Please try to the take the following measures to the original audio data:

- If the sound data has plenty of high frequency sound, it tends to have noise.

  Please reduce high frequency sound using a low pass filter.

- The sound data with harmonics of low or high frequency sound also tends to have noise.

  In this case, you can improve the quality by increasing the sampling frequency of the sound data.

  For instance, enhance the 32kHz data to 64kHz or if you are using 44kHz data, try to convert with AviToMobiclipDs.exe without lowering to 32kHz.

In addition to the above, you also can use an auxiliary audio codec called “FastAudo” in addition to the default audio codec in AviToMobiclipDs.exe. It can provide higher quality sound, but it also has some drawbacks such as slower decoding speed and larger file size. So, please keep these points in mind. For further details of “FastAudio,” please refer the “QuickStart.pdf.”

3 Tips

3.1 Is it possible to play back the movie data from memory?

Q. The function of MO_OpenMovie can only open the ROM file system. Is there any way to playback the movie data from memory?

A. With FS_CreateFileFromMemory, you can playback the movie from the memory.

The function of FS_CreateFileFromMemory enables you to open the designated memory area as a ROM file system. Please give this FSFile structure as an argument to the Mo_OpenMovie function.

For details of FS_CreateFileFromMemory, please refer to the reference manual of NitroSDK.

3.2 Is it possible to do the decode processing within a separate thread?

Q. We are planning to execute the movie decode processing and playback on separate threads. Will there be any problem if we implement this method?

A. No, there will be no problem.

Mobiclip is designed based on the assumption of multi-thread processing. The effectiveUsage sample within the package provides you with samples for such ways of use. Please refer to them.

4 Changes done in new VFW codec using Mobiclip license.

4.1 StarForce protection

The StarForce protection system has been dropped in favor of a simpler protection system based on Mobiclip license files.

A Mobiclip license consists of a file named "User.lic". Please have a look at the "Mobiclip Tools for NINTENDO DS" chapter of QuickStart.pdf document for more details.

4.2 OS compatibility list

The new Mobiclip VFW codec for NINTENDO DS has been successfully tested on the following Windows OSes:

- Windows 7 64 bits
- Windows XP 32 bits with SP3

While not tested it should also work on the following Windows OSes:

- Windows 7 32 bits
- Windows Vista 64 bits
- Windows Vista 32 bits
- Windows XP 64 bits

4.3 VFW codec GUI and feature changes

The new Mobiclip VFW codec for NINTENDO DS has significant differences in term of GUI and features list compared to the old codec using StarForce protection.

This section will list all differences of these two codecs:

- Added "Save Settings" and "Load Settings" buttons that allow to store/load the current codec settings to/from a text file.
- Added 2 encoding types:
  - "Multipass - CBR - First Pass".
  - “Multipass - CBR - Nth Pass".
- Renamed encoding types:
  - From "Multipass - First Pass” to "Multipass - VBR - First Pass".
  - From "Multipass - Next Pass" to "Multipass - VBR - Nth Pass".
- The "Bitrate Curve Compression" parameter was removed of the "Multipass - VBR - Nth Pass" encoding type parameters.
- Renamed "Single Pass" encoding type targeting a quality parameter to "Single Pass - Constant Quality ".
- Renamed "Single Pass" encoding type targeting a bitrate parameter to "Single Pass - CBR ".
- Removed "Single Pass" encoding type targeting a size parameter. This can still be done using "Single Pass - CBR " encoding type by computing needed bitrate to obtain the wanted size by the following fomula:

  Average bitrate (kbps) / 8 * duration (seconds) = video target size (KBytes)
- The "Max Keyframe Interval" parameter is replaced by a "Maximum interval" parameter. This new parameter is available in all encoding types and expects a floating point value in seconds.

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